

1 Kaba Connector Access Control

The Kaba connector for access control connects ZKS hardware by Kaba to HYDRA. The connection is established via the Kaba communication software B-COMM.

There are different types of Kaba access terminals. Depending on the type of terminal, you can process up to 16 readers in one terminal.

1.1 Requirements

The communication software B-COMM needs to be licensed and installed in version 3.14 or higher.

Additionally to the B-COMM data communication, you need the software option *B-Comm Parameter editor*. We also recommend to use the software option *B-COMM User management*. This option offers the possibility to protect the user interface of the *B-COMM (B-COMM GUI)* with a password.

Kaba only supports the badge technologies LEGIC and MIFARE.

If you have terminals in different time zones, you need a HYDRA system for each time zone. The terminals, the connector and the corresponding HYDRA system must operate in the same time zone.

1.2 Supported terminal types

The following terminal types support the connector:

- Kaba access manager 92 90-K5
- B-Net 92 90-2
- B-Net 92 90

The following access readers support the connector:

- Kaba compact reader 91 04-K5
- Kaba compact reader 91 10-K5
- Kaba remote reader 91 15-K5
 - with the Kaba registration units 90 00-K5, 90 01-K5, 90 02-K5, 90 03-K5, 90 04-K5
- B-Net 91 04
- B-Net 91 05
- Bedanet 91 04
- Bedanet 91 05

1.3 Administration

The connector's installation is not included in the HYDRA installation. The connector is installed with B-COMM if required.

In the HYDRA server, the connector runs as service under Windows or as daemon under Linux. The connector is automatically started on starting the server. The connector operates independently, if HYDRA is started or not. It also works offline. If HYDRA is shut down, the connector continues to accept postings (e.g. access logs) and stores this data on the hard drive. Once HYDRA is started, the connector transfers the postings to HYDRA.

1.4 Configurations

The access control configurations are partly made in HYDRA and partly in B-COMM. While access authorizations are defined in HYDRA, the hardware related settings are made via B-COMM. The following chapters describe the configurations and settings in HYDRA.

1.4.1 Access authorizations

In **HYDRA**, access authorizations are configured via [Access profiles](#) and their assignment to badges ([Access profile assignments](#)). For each access profile, you define in the [Access authorizations](#) which [Access groups](#) in combination with a defined [access time model](#) have the authorization to open the [Access points](#) of this access group.

1.4.1.1 Badges and access profile assignments

If you create, modify or delete a [Badge](#) or an [Access profile assignment](#), the modification is communicated promptly to all affected access terminals. It is not necessary to communicate permanently the complete modified data to the terminal as the connector stores the data communicated to the access terminal. The connector only communicates the differences in case of a modification. You can delete and completely reload the data in the terminal using the checkbox *Reload program* in the *Terminal administration* of the [Terminal configuration](#) (go to: *Terminal configuration - Terminal administration - Reload program*).

If an unauthorized badge tries to open an access point, there is no online query to HYDRA asking if the badge is authorized.

If you modify access authorizations due to modified accesses, access time models, opening hours or public holidays, these modifications are identified cyclically and synchronized with the terminal (got to: *Terminal configuration - tab HR functions - Cyclic loading*).



Access terminals of Kaba do not support validity periods for badges. For this reason, the system identifies access authorizations on a daily basis at midnight and communicates possible modifications to the terminals. This might be an employee's first day in the company or an employee who left the company the previous day. If an access terminal was offline for a longer time, the modifications are only communicated to the terminal when it is online again and when the time defined in the field *Cyclic loading* has elapsed (go to: *Terminal configuration* - tab *HR functions* - *Cyclic loading*).

The access terminals of Kaba do not process the times in the [Badge](#) fields *Valid from* and *Valid until*.

1.4.1.2 Access time models

Kaba processes a maximum of 159 [Access time models](#) per access terminal. You can define up to 16 different access time models per access terminal for each badge. You can define up to 7 [Access periods](#) for each access time model.

If one of these limits is exceeded in one terminal, it is communicated to the HYDRA [System logs](#) and you must change the access time models.

If an access time model has more than 7 access periods, the access time model is not transferred to the terminal and the corresponding badges do not get access.

Modifications in the access time models are checked cyclically and transferred to the access terminal (go to: *Terminal configuration* - tab *HR functions* - *Cyclic loading*).



Concerning the assignment of access time models to badges, the connector requires that every *Subterminal* is exactly assigned to a *Level* in B-Comm (*Level* = Number of subterminal). If you have installed an access terminal following the MPDV installation instructions, the *Subterminal* is automatically assigned to a *Level*. You must not change this assignment in B-COMM later on.



Access terminals of Kaba do not support validity periods for access time models. For this reason, the connector reads the current access time models on a daily basis at midnight and transfers any modifications concerning the validity period to the terminals. If an access terminal is offline for a longer time, the modifications are only communicated to the terminal when it is online again and when the time defined in the field *Cyclic loading* in the tab *HR functions* in the *Terminal configuration* has elapsed.

1.4.1.3 Access time models for opening hours

The connector processes the following fields in the [Access periods](#) of the [Access time models](#) that are used for the [Opening hours](#):

Badge required

If this option is checked in an access period, you need a valid badge including access authorization to open the access. If this option is not checked, there is a so called "permanent activation", i.e. the door opener's relay is permanently activated.

Office unlocked

The function "Office unlocked" is **not** available with access terminals of Kaba.

1.4.1.4 Access time models for access authorizations

The connector processes the following fields in the [Access periods](#) of the [Access time models](#) that are used for the [Access authorizations](#):

PIN code required

Once the badge is read, a PIN code might be required. When using access terminals of Kaba, control the entry of the PIN code via the access time model of the badge's access authorizations and not via the opening hours. Therefore, it depends on the previously read badge, if you also have to enter a PIN code.

1.4.1.5 Configuration of PIN code length

The reader's light flashes if you have to enter a PIN code in addition to the badge to open an access. Only after having entered the correct PIN code, the access opens if you have an access authorization.

The number of digits defined for the PIN code is equal for all employees. The setting is made in an [INI configuration](#):

Name:	BCOMMCONNECTOR
Section:	AUTH
Key:	PINLENGTH
Value:	<number of digits for PIN>

By default, the PIN code includes 4 digits.

1.4.1.6 Access authorizations of access profiles

In the [Access authorizations](#) of the [Access profiles](#) the following fields are **not** processed:

Office unlocked

The function "Office unlocked" is **not** available with access terminals of Kaba.

Bag check

The function "Bag check" is **not** available with access terminals of Kaba.

1.4.1.7 Public holidays

The connector transfers the [Public holidays](#) together with the access authorizations to the access terminals. The terminal always receives the following 8 public holidays. The period covered is up to one year.

Access terminals of Kaba only process one public holiday calendar per terminal. As a consequence, the public holidays are identical for all access points of a terminal. The access terminal identifies all public holidays which do not have an entry in the field *Location* or whose location matches the one of the terminal according to the [Terminal configuration](#). The location defined in the [Access group](#) is **not** relevant for access terminals of Kaba.

1.4.2 Hardware related configurations

As far as the hardware related configurations are concerned, the access terminals of Kaba only process few settings of HYDRA. The majority of the configurations are made in the parameter editor of the B-COMM GUI.

The following chapters describe the HYDRA configurations having an effect on the access terminals of Kaba.

1.4.2.1 Terminal

You have to create the Kaba access terminals in the [Terminal configuration](#). The following fields are relevant for these terminals:

Configurations in the tab *General*

Terminal

Terminal number for unique identification.

Active

The connector only processes active terminals.

Location

The access terminal and the corresponding access points identify all [Public holidays](#) which do not have an entry in the field *Location* or whose location matches the one of the terminal.

Type

Enter the terminal type of the Kaba terminal in this field. The connector only processes terminals that are assigned to the type Kaba in the field *Type*.

Terminal class

The terminal class specifies the terminal settings. By default, enter the terminal class "60" for terminals of the type *Kaba 9290*. In case of customer specific processing, it might be necessary to enter a deviating terminal class.

Operated as HYDRA-PZE/ZKS terminal

Check the option *Operated as PZE/ZKS terminal*.

Cycle duration of status messages

Specify the time interval in hours and minutes at which the connector sends status messages to HYDRA.

IP address

The terminal's IP address. For Kaba terminals, the terminal GID and DID must follow the IP address, separated by a semicolon (Example: 192.168.10.213;**0105**).

Company number/system number

The value entered in this field can override the system number that is defined in the HYDRA [Basic settings](#) for individual terminals.

Configurations in the tab *HR functions***Operation mode**

Set the operation mode *Access terminal*.

Cyclic loading

Time interval at which the connector sends modifications to the terminal. The modifications may concern the access points, the access time models, the opening hours and the public holidays. If you check the option *Reload authorizations* in the *Terminal administration* of the *Terminal configuration*, you can define that modifications are transferred to the terminal at the latest after the end of the *Cycle duration of status messages* (if the terminal is online).



The connector promptly reads modifications in the terminal configuration for Kaba terminals. In case of a **new** terminal, you must first define the terminal configuration and the access points. Then you must check the option *Reload program* in the *Terminal administration* of the *Terminal configuration* in order to transfer the configurations to the terminal.

Terminal administration in the toolbar**Terminal from, to**

You can change the *Terminal administration* for one or several terminals.

Activate terminal

Use the checkbox *Activate terminal* to enable or disable in one action the field *Active* for all preselected terminals.

Reboot

If you check the option *Reboot*, all preselected terminals are rebooted. After the next status message, the connector informs the terminal about the reboot. The reboot is therefore executed within the time period entered in the field *Cycle duration of status messages* (if the terminal is online).

Next reboot on

In the field *Next reboot on*, you can enter the point in time of the reboot. Once this point in time is reached, the terminal is only rebooted after having sent the next terminal status.

Delayed by ... minutes

If you select several terminals in the *Terminal configuration* and you activate the field *Delayed by ... minutes*, you can delay the time for the *Next reboot* for each of the terminals by the number of minutes entered.

Reload program

If you check this option, the complete HYDRA configuration is loaded to the terminal. All badges and access authorizations in the terminal are first deleted and then reloaded. During the data transfer, the badges with authorization might not get access for several minutes. In addition, the access terminal is restarted and the readers do not work for several minutes.

Reload authorizations

If you check the option *Reload authorizations*, all modifications to accesses, access time models, opening hours and public holidays are transferred to the terminal at the latest after the end of the time entered in the field *Cycle duration of status messages* (see field *Cyclic loading* in the tab *HR functions* of the application *Terminal configuration*).

1.4.2.2 Access points

You have to create all access points in HYDRA and assign them to an access group. This chapter describes the fields in the Access point which are processed when connected to the connector.



The connector cyclically loads the access points of a terminal (see field *Cyclic loading* in the tab *HR functions* in the *Terminal configuration*). Once the access point is created or modified, the modification is processed at the latest after the time period entered in the [Terminal configuration](#).

Configurations in the tab *Access point*

Active

The connector only processes active access points. E.g., the connector rejects access logs of inactive access points.

Trigger clocking

If you check the option *Trigger clocking*, you log the access as usual and in addition you generate the clocking for the time and attendance product group whenever an employee uses an access. Depending on the access type, the system automatically generates a clocking-in record if it is an entry and a clocking-out record if it is an exit. An automatic status clocking record is generated if it is a passage.

Access type

Specifies whether the access point is an Entrance, Exit, Passage or entrance Without checking the badge number. Kaba terminals do **not** process the setting *Without checking the badge number*.

Access group

The configuration of access authorizations in the [Access profiles](#) and the configuration of the [Opening hours](#) of an access point is made per [Access group](#).

Terminal

Number of the [Terminal](#) the access point is connected to.

Reader

Number of the reader that opens the access point. You must clearly define the combination of terminal and reader in the system. You can also monitor and control an access point without reader (see application case "Access without reader").

Configurations in the tab *Settings*

Report access status

Select the option *Report access status* to define whether or not the access point status is to be sent to the HYDRA server. Possible values are "None" or "Open and closed". Provided that the function "Alarm System" (ZKS-ALS) has been licensed, the selection is extended by "Only alarms" and "All". The options "Open and closed" and "All" increase server communication and should only be selected for entries that are less frequently opened or where the "Open" status is to be monitored via the escalation management function.

Cycle duration for reporting the access status

The current access status is sent to HYDRA, once the specified time has expired.

Reload access configuration

Kaba access terminals do **not** process the field *Reload access configuration* (go to: *Terminal configuration* - tab *HR functions* - *Cyclic Loading*).

Reload authorizations

Kaba access terminals do **not** process this field (go to: *Terminal configuration* - tab *HR functions* - *Cyclic Loading*).

Default Kaba channel assignment

The configuration of the channels (for example for the opener and the status) is not made in HYDRA. If there is no other configuration in B-COMM, use the default Kaba channel assignment.

Reader/Slot	Channel opener	Channel status	Channel open. pushbutton
1	Int O 01	Int I01	Int I02
2	Int O 02	Int I03	Int I04
3	Int O 03	Int I05	Int I06
...	...	...	...
8	Int O 08	Int I15	Int I16
9	Ext O 01	Ext I02	Ext I03
10	Ext O 01	Ext I02	Ext I03
...	...	...	...
16	Ext O 01	Ext I02	Ext I03



The access terminal processes the outputs and inputs of up to 8 readers. '**O**' stands for output and '**I**' for input. From the 9th reader on, the inputs and outputs are processed in the reader.

Configurations in the tab *Advanced Settings*

Entrance to room zone

In this field, define a [room zone](#) that is entered through the access point.

Exit from room zone

In this field, define a room zone that is exited through the access point.

1.4.2.3 Room zones

The connector logs which badges are present in specified [Room zones](#), if you define the fields in the access configuration as *Entrance to room zone* or *Exit from room zone*.

The following fields in the configuration of room zones are **not** processed.

Plausibility check when entering, Plausibility check when exiting

Checking these two options does **not** have any effect on the behavior of the Kaba access terminals. If you want to check, if a badge is already present in a room zone, you can configure this in B-COMM. In this case, all entrances and exits of the room zone must be connected to one access terminal. There is **no** synchronization of room zones configured in HYDRA and room zones configured in B-COMM.

Maximum occupancy, channel for max. occupancy

Kaba access terminals do **not** process these two fields.

1.5 Fields of application

This chapter describes common fields of application and provides further information on the configuration in HYDRA and B-COMM.

1.5.1 One access point with one reader and one door opener key

This is the default setting for new access points. In the B-COMM *Door management*, type 2 "Door with 1 subterminal and 1 door opener key (1 slot)" is set.

If there actually is a door opener key, you must additionally activate the button in B-COMM. The inputs and outputs are assigned by default as described in the table "Default Kaba channel assignment" in the chapter "Accesses".

1.5.2 One access point with 2 readers (inside and outside)

You use this setting, if you use a reader to check inside and outside (for example at a turnstile) which badge is authorized to pass in which direction. What is special in this case is that both readers share one access status contact.

In this case, you must create 2 access points with consecutive reader numbers in HYDRA. Connect both accesses to the same terminal. Enter the same number for both access points in the field *Channel status* to document this case in HYDRA.

In B-COMM, you change to type 3 in the door management: "Door with 2 subterminals and passthrough control (2 slots)".

1.5.3 One access point without reader

The option "one access point without reader" is applied, if you want to monitor the status of an access point or open a door permanently for a specified time, although there is no reader for opening the door.

In HYDRA, you have to create the access point and leave the field *Reader* empty. Instead, you enter the number of the slot (i.e. the number of the reader, if the access point had a reader) in the field *Channel opener* or in the field *Channel status*. Example: If the access point without reader is set to slot 3 in the parameter editor of B-COMM, you have to enter the value 3 in the field *Channel opener* in HYDRA. As an alternative, you can enter the value 3 in the field *Channel status*. According to the default Kaba channel assignment, the status contact must be connected to the input I05 of the access terminal.

In this case, change to type 1 in the B-COMM *Door management*: "Door with 2 door opener keys (1 slot)".

1.5.4 Security gates

A security gate consists of several access points; only one of the access points can be opened at a time.

In HYDRA, you create the access points of the security gate as independent accesses with consecutive reader numbers.

In the B-COMM *Door management*, change to type 4: "Security gate with 2 subterminals and 2 door opener keys (2 slots)"; or change to type 5: "Security gate with 4 subterminals (4 slots)".

1.5.5 Check attendance in room zone

You must activate the setting in B-COMM that a badge can only enter a room zone, if the badge is not already present, or a badge can only leave a room zone, if the badge is present. Please note that the connector requires each reader number to be assigned exactly to a level number in B-COMM. Do not change the assignment in B-COMM.

Room zone settings in HYDRA only refer to the logging of attendance in a room zone. If you control the attendance of a badge in a room zone in HYDRA, you do not trigger a check at the access terminal.

1.5.6 Anti passback

An anti passback is used to prevent an employee from passing their badge after successful entry to a second person who enters with the same badge.

You can configure an anti passback in the B-COMM *Door management* for one access point. For a group of accesses, you configure a collective anti passback in a terminal.

1.5.7 Second channel opener

In B-COMM, you can configure if you want to open two doors with one authorized badge:

1. **I/O Mapping**

In the input/output mapping table, you connect a free logic output (e.g. BO33) to a free physical output (e.g. internal no. 13).

2. **Door management**

In the *Door management* go to tab *Door 1*, group *Passing* and enter the logic output (in the example BO33) in the field *Relay authorized*.

You can also define the time for the relay activation in this field.

You need to restart the terminal to activate these modifications.

It is **not** possible to configure a delay time for the second door in B-COMM.

1.5.8 "Office unlocked" in online components

The option "Office unlocked" offers the possibility to open an access permanently using an authorized badge. The access is closed when the badge is read a second time. This functionality is **not** available with access terminals of Kaba.

1.5.9 Elevator control

The elevator control offers the possibility to control the authorizations for several access points (floors) via a reader. Kaba access terminals do not cover this option by default. If you want to implement an elevator control, you need a customized solution by Kaba, a so-called AVISO routine. Additionally to the AVISO routine, you need the software option AVISO in the terminal according to Kaba's price list. AVISO routines are subject to specific terminals. If you replace the terminal with another one, you must assign the routine to the new terminal.

1.5.10 Bag check

Kaba access terminals do not provide the function [Bag check](#). Use the AVISO routine to implement a customer-specific solution.

1.5.11 Alarm suppression

If an [Alarm suppression](#) is activated in HYDRA, the connector suppresses alarms for the defined period of time. The connector does not transfer the alarms to HYDRA. The alarm suppression has **no** effect on an output that is configured in B-COMM. The alarm is not suppressed.